constituents (rtb cs)

Purpose

`constituents.cs` lists the pesticides, pathogens, heavy metals, salt ions and other constituents that a run simulates. The last group, the other constituents, is the constituent module: selenate (SeO_4), selenite (SeO_3) and boron in soils, aquifers, channels, reservoirs and wetlands. The module simulates sorption, selenium reduction and selenium release from shale, and the constituents move with water like the other dissolved substances.

The salt ion files are described on the salt inputs page. Apart from `constituents.cs`, which is named in `file.cio`, the files are not listed in `file.cio`; SWAT+ looks for them by name in the run directory.

constituents.cs

The master list, named on the simulation line of `file.cio` (default `constituents.cs`). If the file is missing or named `null`, no pesticides, pathogens, salts or other constituents are simulated.

Line 1 is a title. Then come five groups, each as a count followed by that many names:

Group	Count	Names
1	number of pesticides	pesticide names, matched by name to `pesticide.pes`
2	number of pathogens	pathogen names, matched by name to `pathogens.pth`
3	number of heavy metals	metal names, not matched to any database (see metals)
4	number of salt ions	salt ion names, in the order given on the salt inputs page
5	number of other constituents	constituent names

Each count is read from a separate line and the names that follow are read as one list, starting on the next line, so they can be on one line or several. A count of 0 still needs its names line: SWAT+ skips one line after the count, so leave a blank line (or any text) there. A pesticide or pathogen name that is not in its database is kept in the list but gets no parameters.

The constituent module uses the position of each constituent and not its name. Fertilizer input, selenium reactions and the reservoir parameters assume this order:

Position	Constituent
1	`seo4` (selenate)
2	`seo3` (selenite)
3	`boron`

File overview

File	Content	Linked to objects by
`cs_hru.ini`	Initial dissolved and sorbed constituents in HRU soils	`cs` column of `soil_plant.ini`

File	Content	Linked to objects by
cs_aqu.ini	Initial dissolved and sorbed constituents in aquifers	cs column of initial.aqu_cs
cs_channel.ini	Initial concentrations in channels	cs column of initial.cha_cs
cs_res	Reservoir and wetland parameters and initial concentrations	cs column of reservoir.res_cs and wetland.wet_cs
cs_irrigation	Concentrations in irrigation water from outside the model	position, see below
fertilizer.frt_cs	Constituent content of fertilizers and manures	position in the fertilizer list
cs_reactions	Reaction parameters for soils and aquifers	HRU and aquifer order
cs_uptake	Daily constituent uptake by each plant	position in plants.plt
cs_atmo.cli	Concentration in rainfall and dry deposition	atmospheric deposition stations, see climate inputs
cs_recall.rec	Loads from point sources	see recall inputs
cs_plants_boron	Boron tolerance of plants	read but not used
cs_urban	Constituent content of sediment from impervious areas	read but not used
cs_streamobs	Channels for daily concentration output	not read

Missing files

cs_hru.ini, cs_aqu.ini and cs_channel.ini are opened whether or not they exist. If one is missing, SWAT+ creates an empty file of that name in the run directory and reads no records from it. An empty file, whether created this way or supplied, is read the same way.

The test project refdata/Ames_sub1 in the SWAT+ repository contains empty cs_hru.ini, cs_aqu.ini and cs_channel.ini files created by earlier runs. It gives null for the constituent file in file.cio, so no constituents are simulated and the empty files have no effect on the results. If constituents are listed in constituents.cs, an empty file behaves as follows:

- Empty cs_hru.ini: there is no record for the HRUs, but HRU setup still reads record 1, which does not exist. Give at least one record.
- Empty cs_aqu.ini: when initial.aqu_cs is present, the aquifer search runs over the number of records in cs_irrigation and reads cs_aqu.ini records that do not exist. Give at least as many records as cs_irrigation has.
- Empty cs_channel.ini: there is no record for the channels, but channel setup still reads record 1, which does not exist. Give at least one record.

Two files are needed as soon as other constituents are listed in constituents.cs, because they are read for every HRU without a check that they were found:

- cs_irrigation is read for every HRU during setup.
- cs_uptake is read every day for every HRU with a growing plant.

Provide both whenever constituents are simulated, with zero values if there is nothing to add.

cs_hru.ini

Initial constituents in HRU soils. Lines 1 to 5 are skipped. Each record then takes three lines:

Line of record	Content
1	Record name
2	Soil values: one dissolved concentration (mg/L) per constituent, then one sorbed concentration (mg/kg of soil) per constituent

Line of record	Content
3	Plant values: the same number of values, read but not used

An HRU uses the record named in the `cs` column of its `soil_plant.ini` record; if the name is not found, the first record is used. `soil_plant.ini` has the `cs` column only when `event` in `codes.bsn` is not 0. With `event` at 0, the column is not read and every HRU uses the first record of `cs_hru.ini`. Every soil layer starts with the same values.

cs_aqu.ini

Initial constituents in aquifers. Lines 1 to 3 are skipped. Each later line is one record: name, one dissolved concentration (mg/L) per constituent, then one sorbed concentration (mg/kg) per constituent. The sorbed mass is calculated with an assumed aquifer bulk density of 2000 kg/m³.

Aquifers find their record through `initial.aqu_cs` (title, header, then rows with `name`, `pest`, `path`, `hmet`, `salt`, `cs`). Aquifer n uses row n of `initial.aqu_cs`, whatever the row's name, and the `cs` column of that row names a record in `cs_aqu.ini`. See salt inputs for the same link used by salts.

cs_channel.ini

Initial concentrations in channel water. Lines 1 and 2 are skipped. Each later line is one record: name and one concentration (mg/L) per constituent. Channels find their record through `initial.cha_cs` as described for `salt_channel.ini`, using its `cs` column.

cs_res

Constituent parameters and initial concentrations for reservoirs and wetlands. Lines 1 to 15 are skipped (two title lines, twelve comment lines and a header). The file is ignored when the reservoir nutrient file in `file.cio` is `null`. Each later line is one record:

#	Column	Units	Default	Description
1	`name`	-	-	Record name, named in the `cs` column of `reservoir.res_cs` or `wetland.wet_cs`.
2	`v_seo4`	m/day	0.001	Settling velocity of selenate.
3	`v_seo3`	m/day	0.001	Settling velocity of selenite.
4	`v_born`	m/day	0.001	Settling velocity of boron.
5	`k_seo4`	1/day	0.05	First-order loss rate of selenate.
6	`k_seo3`	1/day	0.03	First-order loss rate of selenite.
7	`k_born`	1/day	0.0	First-order loss rate of boron.
8	`theta_seo4`	-	1.08	Temperature factor for the selenate rate.
9	`theta_seo3`	-	1.08	Temperature factor for the selenite rate.
10	`theta_born`	-	1.08	Temperature factor for the boron rate.
11	`c_seo4`	mg/L	0.0	Initial selenate concentration.
12	`c_seo3`	mg/L	0.0	Initial selenite concentration.
13	`c_born`	mg/L	0.0	Initial boron concentration.

All 13 values must be given; a short line takes its remaining values from the next line. A reservoir without a `cs_res` record starts with no constituents and uses the default rates in the table. `reservoir.res_cs` and `wetland.wet_cs` are described under `salt_res`.

cs_irrigation

Concentrations in irrigation water from outside the model. Lines 1 and 2 are skipped. Each later line is one record: name and one concentration (mg/L) per constituent.

An HRU uses the record at the same position as the `cs_hru.ini` record it uses; the name is not used. In 62.0.1 the transfer of constituents with irrigation water is switched off in water allocation, so these concentrations do not change the constituent balance.

fertilizer.frt_cs

Constituent content of each fertilizer. Line 1 is a title and line 2 a header. Then one row per record of `fertilizer.frt`, in the same order and with the same number of rows:

#	Column	Units	Description
1	`name`	-	Fertilizer name. Not checked against `fertilizer.frt`.
2	`seo4`	kg/kg	Selenate per kg of material applied.
3	`seo3`	kg/kg	Selenite per kg of material applied.
4	`boron`	kg/kg	Boron per kg of material applied.

The mass added (kg/ha) is the amount applied (kg/ha) times the value in the table. Constituents are added in the same cases as salts (see `salt_fertilizer.frt`): with manure operations and decision-table `manure` actions whose manure has an `org_min` entry other than the first record of `manure_om.frt`, and with fertilizer operations on HRUs with standing water. For manure, the whole amount goes into the top soil layer.

cs_reactions

Reaction parameters for soils and aquifers, organised in reaction groups and shale formations. Every HRU and every aquifer must have a row.

Section	Content
Line 1, 2	Title and header, skipped
Line 3	Number of parameters n_p and number of reaction groups n_g
next n_p lines	One parameter per line, with one value per reaction group
next line	Header, skipped
next line	Number of shale formations n_s
next n_s lines	Per shale formation: S/Se ratio of the shale, rate constant for autotrophic reduction of oxygen (1/day), rate constant for autotrophic reduction of nitrate (1/day)
next line	Header, skipped
next lines	One line per HRU, in HRU order: HRU number, reaction group, n_s shale fractions (fraction of the HRU area over each shale formation)
next line	Header, skipped (only when the model has aquifers)
next lines	One line per aquifer, in aquifer order: aquifer number, reaction group, n_s shale fractions

The parameter lines are used in this order:

Line	Parameter	Units	Used for
1	Selenium reduction inhibition factor for nitrate	-	soils, aquifers
2	Oxygen concentration in soil water	mg/L	soils
3	Oxygen concentration in groundwater	mg/L	aquifers
4	Sorption coefficient K_d of selenate	L/kg	soils, aquifers
5	Sorption coefficient K_d of selenite	L/kg	soils, aquifers
6	Sorption coefficient K_d of boron	L/kg	soils, aquifers
7	Rate constant for selenate reduction in soil	1/day	soils
8	Rate constant for selenate reduction in groundwater	1/day	aquifers

The first value of the HRU and aquifer lines is not checked; the line position decides. Further parameter lines are read but not used by soils and aquifers. The rate constant for selenite reduction is not read for soils or aquifers and stays 0, so selenite is not reduced there.

cs_uptake

Specified daily constituent uptake by plants. Lines 1 to 3 are skipped. Then one row per plant in plants.plt, in the same order: name and one uptake value (kg/ha per day) per constituent. The plant name is not checked. While a plant has roots, its daily uptake of each constituent is split between the soil layers in proportion to the root mass in each layer and is limited to the mass in the layer.

cs_plants_boron

Boron tolerance of plants. Line 1 is a title and line 2 is skipped, line 3 is a switch (0 = off, 1 = on), lines 4 to 7 are skipped, and then one row per plant in plants.plt: name, threshold a and slope b. In 62.0.1 the values are read but no routine uses them, so boron has no effect on plant growth.

cs_urban

Constituent content of sediment washed from impervious areas. Lines 1 and 2 are skipped. Each later line is an urban type name from urban.urb and one concentration (mg per kg of sediment) per constituent. In 62.0.1 the values are read but not used.

cs_streamobs

A list of channels for daily concentration and load output to `cs_streamobs_output`: a title line, the number of channels, then one channel number per line. In 62.0.1 the switch that would read it is never set, so the file is not read and the output is not written.

Example

`constituents.cs` for a run with one pesticide, the eight salt ions and the three constituents:

```
constituents.cs: constituent list
1
atrazine
0

0

8
so4 ca mg na k cl co3 hco3
3
seo4 seo3 boron
```

There are no pathogens or heavy metals, so their counts are 0 and each is followed by an empty names line.

A matching `cs_aqu.ini` record with 0.005 mg/L selenate in groundwater and 0.2 mg/kg sorbed selenate, values illustrative:

```
cs_aqu.ini: initial constituents in aquifers
seo4 seo3 boron: dissolved (mg/L), then sorbed (mg/kg)
name          seo4     seo3    boron   seo4_s   seo3_s  boron_s
aqu_cs1      0.005    0.001    0.100    0.200    0.050    1.000
```

Related files

- salt inputs: the salt ions listed in `constituents.cs`, and the link files `initial.aqu_cs`, `initial.cha_cs`, `reservoir.res_cs` and `wetland.wet_cs`.
- file.cio: names `constituents.cs`.
- soil_plant.ini: names the `cs_hru.ini` record for each HRU.
- codes.bsn: `event` decides whether soil_plant.ini has the `cs` column.
- fertilizer.frt: sets the row order of `fertilizer.frt_cs`.
- pesticide.pes, pathogens.pth, metals: the other groups in `constituents.cs`.

- cs_atmo.cli and cs_recall.rec: deposition and point-source loads.
- gwflow inputs: constituents become groundwater solutes when gwflow solute transport is on.

Sources

- Master list: constit_db_read.f90; type cs_db_type in constituent_mass_module.f90
- Readers: cs_hru_read.f90, cs_aqu_read.f90, cs_cha_read.f90, res_read_csdb.f90, cs_irr_read.f90, cs_fert_read.f90, cs_reactions_read.f90, cs_uptake_read.f90, cs_plant_read.f90, cs_urban_read.f90
- Link files and initialisation: soil_plant_init.f90, cs_hru_init.f90, aqu_read_init_cs.f90, sd_channel_read.f90, sd_hydsed_init.f90, res_read_salt_cs.f90
- Processes: cs_fert.f90, cs_fert_wet.f90, cs_irrig.f90, cs_uptake.f90 (called from pl_biomass_gro.f90), cs_sorb_hru.f90, cs_sorb_aqu.f90, se_reactions_soil.f90, se_reactions_aquifer.f90, res_cs.f90
- Types and variables: cs_module.f90, cs_data_module.f90, res_cs_module.f90

File and column	Fortran variable
constituents.cs counts	cs_db%num_pests, %num_paths, %num_metals, %num_salts, %num_cs
constituents.cs names	cs_db%pests, %paths, %metals, %salts, %cs
cs_hru.ini soil and plant lines	cs_soil_ini%soil, cs_soil_ini%plt
cs_aqu.ini	cs_aqu_ini%aqu
cs_channel.ini	cs_cha_ini%conc
cs_res	res_cs_data (type reservoir_cs_data)
cs_irrigation	cs_water_irr%water, copied to cs_irr%csc
fertilizer.frt_cs	fert_cs (type fert_db_cs)
cs_reactions parameter lines 1 to 8	rct(1:8, group), copied to se_ino3, oxy_soil, oxy_aqu, kd_seo4, kd_seo3, kd_born, kseo4 of cs_rct_soil and cs_rct_aqu
cs_reactions shale lines	rct_shale(:,1:3), copied to sseratio, ko2a, kno3a
cs_uptake	cs_uptake_kg
cs_plants_boron	bor_tol_sim, bor_stress_a, bor_stress_b
cs_urban	cs_urban_conc
cs_streamobs	cs_str_obs